Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		Substitution: $\frac{1}{R_{\text{total}}} = \frac{1}{10} + \frac{1}{10}$ (1) Total resistance = 5 [Ω] (1) Award 1 mark for answer only of 0.2 [Ω] **Alternative:** $\frac{100}{20}$ (1) = 5 [Ω] (1)	1	1		2	2	2
		(ii)		Substitution: $I = \frac{V}{R} = \frac{12}{5}$ (1) **ecf** I = 2.4 [A] (1) **Alternative:** $I = \frac{12}{10}$ = 1.2 [A] (1) × 2 = 2.4 [A] (1)	1	1		2	2	2
		(iii)		12 [V]	1			1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iv)		Substitution: $I = \frac{V}{R} = \frac{12\ \mathbf{ecf}}{10}$ (1) I = 1.2 [A] (1) **Alternative:** $I = \frac{2.4\ \mathbf{ecf}}{2}$ (1) I = 1.2 [A] (1) N.B. Treat answer of 1 A as neutral	1	1		2	2	2
	(b)	(i)		Increases		1		1		1
		(ii)		Decreases		1		1		1
		(iii)	I	Decreases		1		1		1
			II	Voltage is shared in series **or** each gets 6 V		1		1		1
				Question 3 total	**4**	**7**	**0**	**11**	**6**	**11**